Effective Models of Traversable Wormhole Mouths: Geometry, Fields, Response, and Temporal Holonomy

Jan Olšina, AI generated paper

Independent Researcher, Prague, Czech Republic
email: jan.olsina@gmail.com, ORCID: 0009-0003-9816-3958.

***Abstract*—This note develops a hierarchy of effective models for a traversable wormhole with two compact mouths. It does not propose a new exact solution of the Einstein equations. Its purpose is to separate consequences of the mouth identification from additional assumptions about field matching, homogeneous topological sectors, and the exotic system that supports the throat. A zero-length spherical model is used to formulate traversal kinematics. The differential of the boundary identification fixes tangential motion, whereas a complete four-momentum map also requires a prescription for the transverse throat direction; an ideal elastic prescription reverses the mouth-normal component and applies a Lorentz transformation between mouth frames. Static scalar fields obey their usual local equations away from the throat, supplemented by pointwise potential and normal-flux matching in a chosen global-potential sector. For two flat exteriors glued at a sphere, the electrostatic response of a point charge is derived explicitly. Unlike the Kelvin image of a grounded conductor, the same-side wormhole field is represented by a uniform line image ending at the Kelvin point, up to a separately specified source-free flux mode. The Newtonian limit of general relativity is then used to interpret local mass-type monopoles near the mouths. In the ideal short-throat, fixed-total-monopole convention, unequal external potentials produce equal-and-opposite monopole coefficients, but this is an effective boundary-model result rather than a universal passive material response. Finally, accelerated mouths and a final constant clock offset are described by a time translation in the gluing map. A smooth Ellis-type throat shows that the local coordinate transition associated with a prescribed constant offset adds neither curvature nor a force on structureless test particles. The offset becomes invariant temporal holonomy only in the completed multiply connected spacetime, where it can affect waves, self-interaction, and global causality.**

***Index Terms*—traversable wormholes, boundary identification, image charges, self-force, Newtonian limit, thin shells, temporal holonomy, time machines, effective models.**

I. Scope and Hierarchy of Assumptions

This document is an exploratory effective-model note. It is not a construction theorem for traversable wormholes, and it does not establish that the assumed mouth geometries can be supported by physically reasonable matter. Under the standard assumptions of classical general relativity, static traversable throats require violations of the null energy condition at or near the throat [1], [3]. Thin-shell models make the required surface stress explicit through the junction conditions [4], [5].

The analysis will keep five logically distinct ingredients separate:

1) the exterior geometry and topology;
2) the identification of corresponding points on the two mouth worldtubes;
3) the local equations obeyed by matter and fields away from the throat;
4) matching conditions and homogeneous topological field sectors;
5) the constitutive response of the throat-supporting system.

The first three ingredients determine much of the kinematics. The fourth fixes the boundary-value problem for a chosen field sector. The fifth is needed for stability, recoil, energy storage, deformation, and the dynamical response of the supporting matter.

Throughout the note, the following terminology is used.

- A *geometric consequence* follows from the quotient geometry or from a local field equation.
- A *matching assumption* specifies how fields or tangent vectors continue through the throat.
- A *homogeneous-mode convention* fixes source-free flux or a common asymptotic monopole that is not determined by local sources.
- A *constitutive model* specifies the response and stored energy of the throat-supporting system.

This hierarchy is important because several simple formulas below are exact within a stated boundary model but are not universal predictions of general relativity.

II. Mouth Geometry and Identification

A. Spherical zero-length model

Let two spherical mouths of radius a have centers $\mathbf{X}_A(t)$ and $\mathbf{X}_B(t)$, with body-frame orientations $R_A(t), R_B(t) \in SO(3)$. Remove the interiors of the two spheres from the exterior space. A boundary point on mouth A is

$$\mathbf{x}_A = \mathbf{X}_A + aR_A\mathbf{n}, \qquad |\mathbf{n}| = 1. \tag{1}$$

The corresponding point on mouth B is

$$\mathbf{x}_B = \mathbf{X}_B + aR_BP\mathbf{n}, \tag{2}$$

where $P \in O(3)$ specifies the internal angular identification. Define

$$\mathcal{O} = R_BPR_A^T. \tag{3}$$

The radial unit normals pointing from the mouth centers into their respective exteriors are

$$\boldsymbol{\nu}_A = R_A\mathbf{n}, \qquad \boldsymbol{\nu}_B = R_B P\mathbf{n} = \mathcal{O}\boldsymbol{\nu}_A. \tag{4}$$

A spacetime identification can be written

$$(\tau, \mathbf{n})_A \sim (\tau, P\mathbf{n})_B, \tag{5}$$

where τ is an internal throat-time label. Equation (5) is an equivalence relation. Before the quotient, its two sides are distinct boundary events; after the quotient, they represent one event in the completed spacetime. A traversing worldline may therefore appear discontinuous in one exterior chart while remaining continuous in an atlas adapted to the quotient.

The differential of the boundary map acts only on vectors tangent to the mouth. It does not, by itself, determine how a vector is continued in the transverse direction. A full traversal rule must therefore specify an extension from the boundary tangent bundle to the normal bundle.

B. *Finite throat*

A smooth static spherical throat may be written in proper longitudinal coordinate l as

$$\mathrm{d}s^2 = -N^2(l)c^2\mathrm{d}\tau^2 + \mathrm{d}l^2 + r^2(l)\mathrm{d}\Omega^2, \tag{6}$$

where $r(l)$ has a positive minimum. Exterior regions can be attached at two values of l, or may arise as the two asymptotic ends of the same smooth solution. The zero-length model is a boundary-theory idealization of such a geometry, not its unique regularization.

III. Traversal Kinematics

A. *Normal reversal and the ideal elastic extension*

Let an incoming spatial momentum at mouth A be decomposed as

$$\mathbf{p}_{\text{in}} = \mathbf{p}_{\parallel} + p_{\perp}\boldsymbol{\nu}_A, \qquad \mathbf{p}_{\parallel}\cdot\boldsymbol{\nu}_A = 0. \tag{7}$$

A trajectory approaching the mouth from its exterior has $p_\perp < 0$. To emerge into the destination exterior, its outgoing normal component relative to $\boldsymbol{\nu}_B$ must be positive. The simplest elastic extension of the boundary differential is therefore

$$J(\mathbf{n}) = \mathcal{O}\left(I - 2\boldsymbol{\nu}_A\boldsymbol{\nu}_A^T\right). \tag{8}$$

It satisfies

$$J\boldsymbol{\nu}_A = -\boldsymbol{\nu}_B, \qquad J\mathbf{v}_{\parallel} = \mathcal{O}\mathbf{v}_{\parallel}. \tag{9}$$

Thus the tangential components follow the differential of the mouth identification, while the transverse component reverses. This normal reversal is part of the chosen ideal traversal prescription; it is not fixed by the boundary map alone.

B. *Four-momentum map*

Let ${e_A}^{\hat{a}}{}_{\mu}$ and ${e_B}^{\mu}{}_{\hat{a}}$ be orthonormal tetrads at the entrance and exit events. A general idealized traversal law can be represented by a Lorentz transformation $W^{\hat{a}}{}_{\hat{b}}$:

$$p^{\mu}_{\text{out}} = {e_B}^{\mu}{}_{\hat{a}} W^{\hat{a}}{}_{\hat{b}} {e_A}^{\hat{b}}{}_{\nu} p^{\nu}_{\text{in}}. \tag{10}$$

For static synchronized mouths in a common approximately inertial frame, the ideal elastic choice is

$$W = \begin{pmatrix} 1 & 0 \\ 0 & J(\mathbf{n}) \end{pmatrix}. \tag{11}$$

Because this W is Lorentz,

$$p^{\mu}_{\text{out}} p_{\mu}^{\text{out}} = p^{\mu}_{\text{in}} p_{\mu}^{\text{in}}. \tag{12}$$

The mass shell is therefore preserved by assumption in the ideal elastic model. This is not a consequence of topology alone; an inelastic or radiative throat could obey a different effective map.

C. *Momentum bookkeeping*

Four-momenta at separated events in a curved spacetime cannot be subtracted without a transport prescription. In a fixed asymptotically inertial frame, however, and neglecting radiation or momentum exchanged with external supports during the crossing interval, the net change assigned to the wormhole-plus-support system is approximately

$$\Delta P^{\mu}_{\text{wh}} \simeq p^{\mu}_{\text{in}} - p^{\mu}_{\text{out}}. \tag{13}$$

A useful local bookkeeping split is

$$\Delta P^{\mu}_A \simeq p^{\mu}_{\text{in}}, \qquad \Delta P^{\mu}_B \simeq -p^{\mu}_{\text{out}}. \tag{14}$$

Only the total asymptotic balance is invariant under changes in how recoil, shell stress, radiation, and actuator momentum are partitioned.

If a spatial map rotates a momentum of magnitude p through an angle α, then

$$|\Delta\mathbf{P}_{\text{wh}}| = 2p\sin\left(\frac{\alpha}{2}\right). \tag{15}$$

A reversal gives $2p$. Identical asymptotic input and output momenta give no net impulse to the pair, although the two mouths can still receive opposite local impulses.

D. *Moving and rotating mouths*

In the nonrelativistic limit, let the surface velocity at a crossing point be

$$\mathbf{v}_i^{\text{surf}} = \mathbf{V}_i + \boldsymbol{\omega}_i \times \mathbf{r}_i. \tag{16}$$

The elastic velocity rule corresponding to (8) is

$$\mathbf{u}_{\text{out}} = \mathbf{v}_B^{\text{surf}} + J(\mathbf{n})\left(\mathbf{u}_{\text{in}} - \mathbf{v}_A^{\text{surf}}\right). \tag{17}$$

The kinetic-energy change of a particle of mass m is

$$\Delta E_{\text{particle}} = \frac{m}{2}\left(|\mathbf{u}_{\text{out}}|^2 - |\mathbf{u}_{\text{in}}|^2\right). \tag{18}$$

Any gain is supplied by mouth motion, an actuator, or an internal reservoir. Equation (17) is a nonrelativistic approximation to composing the two mouth boosts with the internal Lorentz map W.

IV. Local Field Equations and Throat Matching

A. Effective action

A broad relativistic action is

$$S = \frac{c^3}{16\pi G}\int R\sqrt{-g}\,\mathrm{d}^4x - \frac{1}{4\mu_0}\int F_{\mu\nu}F^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x + S_{\mathrm{support}} + S_{\mathrm{matter}}. \tag{19}$$

The unknown term S_{support} maintains the throat and determines its mechanical and field response.

For quasistatic scalar fields in an exterior spatial domain $\mathcal{V}$, a convenient weak-field functional is

$$\begin{aligned} L_{\mathrm{fields}} = &\int_{\mathcal{V}} \left[\frac{\epsilon_0}{2}|\boldsymbol{\nabla}\phi|^2 - \rho_e\phi\right]\mathrm{d}^3x \\ &+ \int_{\mathcal{V}} \left[-\frac{1}{8\pi G}|\boldsymbol{\nabla}\Phi|^2 - \rho_m\Phi\right]\mathrm{d}^3x. \end{aligned} \tag{20}$$

The source densities may include point-particle delta functions. Variation gives

$$\nabla^2\phi = -\frac{\rho_e}{\epsilon_0}, \qquad \nabla^2\Phi = 4\pi G\rho_m. \tag{21}$$

The topology does not alter these local equations away from the throat. It alters their global solution space and boundary matching.

B. Potential and flux matching

Assume a static gauge in which a single-valued scalar potential exists on the completed spacetime, with no voltage layer or nontrivial gauge transition inserted at the throat. Then corresponding points obey

$$\phi_A(\mathbf{n}) = \phi_B(P\mathbf{n}). \tag{22}$$

Let ∂_{ν_i} denote differentiation in the radial direction from mouth i into its exterior. Smooth continuation through a short throat gives

$$\partial_{\nu_A}\phi_A(\mathbf{n}) = -\partial_{\nu_B}\phi_B(P\mathbf{n}), \tag{23}$$

when no surface charge is inserted at the gluing surface. The minus sign reflects the reversal between the two exterior radial directions.

The same local equations can admit source-free harmonic fields. In the simplest two-ended electrostatic problem, these are characterized by flux through the throat rather than by independent higher multipoles [9]. Fixing both asymptotic potentials to the same value removes this homogeneous flux mode. Allowing different asymptotic potential constants permits an additional source-free throat flux. The charge-generated field and the source-free field must therefore be specified separately.

C. Topological self-interaction

For a static Green function, write

$$G_{\mathrm{wh}}(\mathbf{x},\mathbf{x}') = G_0(\mathbf{x},\mathbf{x}') + G_{\mathrm{top}}(\mathbf{x},\mathbf{x}'), \tag{24}$$

where G_0 contains the local Coulomb singularity. After regularization, a charge has self-energy

$$U_{\mathrm{self}}(\mathbf{x}) = \frac{q^2}{2}G_{\mathrm{top}}(\mathbf{x},\mathbf{x}) \tag{25}$$

under the convention used to normalize the Green function, and self-force

$$\mathbf{F}_{\mathrm{self}} = -\boldsymbol{\nabla}U_{\mathrm{self}}. \tag{26}$$

Explicit calculations for symmetric wormhole profiles find topology-dependent self-forces. Their detailed expressions depend on the geometry and on the treatment of the source-free flux sector; in the models of Khusnutdinov and Bakhmatov the force is attractive, while Krasnikov emphasizes a different source-generated/source-free decomposition and obtains a different short-throat result [9], [10].

V. Spherical Electrostatic Boundary Problem

A. Conducting-sphere benchmark

A point charge q at $\mathbf{x}_0 = D\hat{\mathbf{n}}$, with $D > a$, outside a grounded conducting sphere of radius a is represented by the Kelvin image

$$\mathbf{x}' = \frac{a^2}{D}\hat{\mathbf{n}}, \qquad q' = -q\frac{a}{D}. \tag{27}$$

The exterior potential is

$$\phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{|\mathbf{x}-\mathbf{x}_0|} + \frac{q'}{|\mathbf{x}-\mathbf{x}'|}\right], \tag{28}$$

with induced surface density

$$\sigma(\theta) = -\frac{q}{4\pi a}\frac{D^2-a^2}{(D^2+a^2-2aD\cos\theta)^{3/2}}. \tag{29}$$

An isolated neutral sphere requires an additional central image that restores zero total induced charge [14], [15].

For two equal conducting spheres in a symmetric source-free environment, equal conductor potentials imply equal charges by exchange symmetry. In a general asymmetric external field, however, the relation is

$$V_i = V_i^{\mathrm{ext}} + \sum_j P_{ij}Q_j, \tag{30}$$

so equal total potentials need not imply equal induced charges. This distinction matters because a conductor enforces one constant potential over an entire connected surface, whereas a wormhole imposes pointwise matching only between corresponding mouth points.

B. Two flat exteriors glued at a sphere

Consider two copies of $r > a$, denoted $+$ and $-$, whose spheres at $r = a$ are identified. A point charge q is in the $+$ exterior at $r_0 > a$. Assume:

1) both asymptotic potentials vanish;
2) the potentials are continuous at corresponding points;
3) their radial derivatives into the two exteriors are opposite;
4) no additional source-free throat-flux mode is added.

For an axis through the charge, expand

$$4\pi\epsilon_0\phi_+ = \frac{q}{|\mathbf{x}-\mathbf{x}_0|} + \sum_{\ell=0}^{\infty}\frac{A_\ell}{r^{\ell+1}}P_\ell(\cos\gamma), \tag{31}$$

$$4\pi\epsilon_0\phi_- = \sum_{\ell=0}^{\infty} \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos\gamma). \tag{32}$$

At $r = a < r_0$,

$$\frac{q}{|\mathbf{x} - \mathbf{x}_0|} = q \sum_{\ell=0}^{\infty} \frac{a^\ell}{r_0^{\ell+1}} P_\ell(\cos\gamma). \tag{33}$$

Potential continuity gives

$$B_\ell = q\frac{a^{2\ell+1}}{r_0^{\ell+1}} + A_\ell. \tag{34}$$

Using $\partial_r\phi_+ = -\partial_r\phi_-$ at the two radial exteriors then gives

$$A_\ell = -\frac{q}{2(\ell+1)} \frac{a^{2\ell+1}}{r_0^{\ell+1}}, \tag{35}$$

$$B_\ell = \frac{q(2\ell+1)}{2(\ell+1)} \frac{a^{2\ell+1}}{r_0^{\ell+1}}. \tag{36}$$

The result differs from the grounded conductor because the wormhole imposes coupled continuity and flux conditions rather than a Dirichlet condition on one surface.

C. Uniform line-image representation

Define the Kelvin radius

$$b = \frac{a^2}{r_0}. \tag{37}$$

The same-side coefficients (35) are generated by a uniform fictitious line charge

$$\mathbf{x}_{\rm im}(s) = s\hat{\mathbf{n}}_0, \qquad 0 \le s \le b, \tag{38}$$

with density

$$\lambda = -\frac{q}{2a}. \tag{39}$$

Indeed,

$$\int_0^b \frac{\lambda\, \mathrm{d}s}{|\mathbf{x} - s\hat{\mathbf{n}}_0|} = -\frac{q}{2a} \sum_{\ell=0}^{\infty} \frac{b^{\ell+1}}{(\ell+1)r^{\ell+1}} P_\ell(\cos\gamma). \tag{40}$$

The total line-image charge is

$$Q_{\rm line} = -\frac{qa}{2r_0}. \tag{41}$$

Thus the Kelvin point remains geometrically distinguished, but it is the endpoint of a distributed image rather than the location of a single point image. A different homogeneous flux convention can add a monopolar source-free contribution together with a relative asymptotic potential offset.

D. Self-energy in the fixed-flux convention

At the source position,

$$\int_0^b \frac{\mathrm{d}s}{r_0 - s} = -\ln\left(1 - \frac{a^2}{r_0^2}\right). \tag{42}$$

The regular potential generated by the line image is therefore

$$\phi_{\rm reg}(r_0) = \frac{q}{8\pi\epsilon_0 a} \ln\left(1 - \frac{a^2}{r_0^2}\right), \tag{43}$$

and the corresponding self-energy is

$$\boxed{U_{\rm self}(r_0) = \frac{q^2}{16\pi\epsilon_0 a} \ln\left(1 - \frac{a^2}{r_0^2}\right).} \tag{44}$$

The radial force is

$$\boxed{F_r = -\frac{q^2 a}{8\pi\epsilon_0 r_0 (r_0^2 - a^2)}.} \tag{45}$$

It points toward the ideal mouth and diverges as $r_0 \to a^+$. The divergence reflects the sharp zero-length boundary idealization. A smooth throat removes the distributional mouth, although a nonzero topology-sensitive self-force generally remains. Equations (44) and (45) apply only to the homogeneous-mode convention stated above.

VI. Weak-Field Gravitational Interpretation

A. Newtonian limit of the exterior metric

In a weak exterior region, write

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1. \tag{46}$$

With the trace-reversed perturbation

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h \tag{47}$$

and harmonic gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$, the linearized Einstein equations are

$$\Box\bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}. \tag{48}$$

For a stationary nonrelativistic source, $T_{00} \simeq \rho c^2$. Defining

$$\bar{h}_{00} = -\frac{4\Phi}{c^2} \tag{49}$$

gives

$$\boxed{\nabla^2\Phi = 4\pi G\rho.} \tag{50}$$

When exterior anisotropic stress is negligible,

$$\begin{aligned} \mathrm{d}s^2 = &- \left(1 + \frac{2\Phi}{c^2}\right) c^2 \mathrm{d}t^2 \\ &+ \left(1 - \frac{2\Phi}{c^2}\right) \delta_{ij} \mathrm{d}x^i \mathrm{d}x^j + O(c^{-4}). \end{aligned} \tag{51}$$

The slow-particle geodesic equation then gives

$$\frac{\mathrm{d}^2\mathbf{x}}{\mathrm{d}t^2} = -\boldsymbol{\nabla}\Phi. \tag{52}$$

B. Lapse matching and its limits

For a static mouth, the lapse is

$$N_i = 1 + \frac{\Phi_i}{c^2} + O(c^{-4}). \tag{53}$$

If corresponding mouth events use the same internal time label and no additional redshift layer or time rescaling is inserted in the gluing map, continuity of the pulled-back induced metric requires

$$N_A(\mathbf{n}) = N_B(P\mathbf{n}). \tag{54}$$

At leading order,

$$\Phi_A(\mathbf{n}) = \Phi_B(P\mathbf{n}). \tag{55}$$

This is a conditional matching statement. A different identification of the mouth time parameters, a finite internal redshift profile, or an explicit shell layer can modify the relation. Equality of the lapse does not determine its normal derivatives; those are related to the supporting stress and other gravitational degrees of freedom.

C. Local mass-type monopoles

In a vacuum buffer region around a compact nearly spherical mouth i, write

$$\begin{aligned}\Phi_i = U_i - \frac{G\delta M_i}{r_i} + \Phi_i^{\mathrm{tidal}} \\ + \sum_{\ell\geq 1}\sum_m \frac{\Phi_{i,\ell m}}{r_i^{\ell+1}} Y_{\ell m}(\theta_i, \varphi_i).\end{aligned} \tag{56}$$

Here U_i is the regular external potential at the mouth center, excluding the mouth's own response, and δM_i is a local mass-type monopole coefficient. In a geometry possessing a separate asymptotic end, an asymptotic $1/r$ coefficient may be related to the ADM mass of that end. For two mouths in one common exterior, however, the individual local coefficients near the mouths are not separately ADM masses; ADM mass is defined at spatial infinity. A Komar interpretation additionally requires stationarity and an appropriate Killing field [5], [8].

D. Ideal spherical response and homogeneous mode

Let two nearly spherical mouths of radius a be separated by $L \gg a$. Retain only the regular external potentials U_A, U_B and local monopoles $\delta M_A, \delta M_B$. To monopole order, lapse matching gives

$$U_A - \frac{G\delta M_A}{a} - \frac{G\delta M_B}{L} = U_B - \frac{G\delta M_B}{a} - \frac{G\delta M_A}{L}. \tag{57}$$

Equivalently,

$$\delta M_A - \delta M_B = \frac{U_A - U_B}{G(a^{-1} - L^{-1})}. \tag{58}$$

The matching condition fixes the antisymmetric monopole in this ideal boundary model but not the common mode $\delta M_A + \delta M_B$. The latter represents additional asymptotic mass, stored throat energy, or a homogeneous response and must be fixed by a separate convention or constitutive law.

Under the fixed-total-monopole convention

$$\delta M_A + \delta M_B = 0, \tag{59}$$

one obtains

$$\boxed{\delta M_A = \frac{U_A - U_B}{2G(a^{-1} - L^{-1})}, \qquad \delta M_B = -\delta M_A.} \tag{60}$$

For $L \gg a$,

$$\delta M_A \simeq \frac{a}{2G}(U_A - U_B). \tag{61}$$

Equation (60) is exact only within the stated monopole truncation, lapse-matching prescription, and common-mode convention.

A broader effective model may be parameterized by

$$\delta M_- \equiv \frac{\delta M_A - \delta M_B}{2} = \chi_0(U_A - U_B), \tag{62}$$

where χ_0 summarizes finite-throat geometry, redshift layers, higher multipoles, support dynamics, and control laws. The ideal short spherical model has

$$\chi_0^{\mathrm{ideal}} = \frac{1}{2G(a^{-1} - L^{-1})}. \tag{63}$$

E. Examples

If a point mass M is outside both mouth spheres and is at equal distance $D > a$ from their centers, the mean-value property for harmonic functions gives the same spherical average of its potential on both mouths:

$$U_A = U_B = -\frac{GM}{D}. \tag{64}$$

The ideal antisymmetric monopole then vanishes, although higher multipoles can remain if the angular patterns differ.

If the source is at distances $D_A, D_B > a$,

$$U_A = -\frac{GM}{D_A}, \qquad U_B = -\frac{GM}{D_B}. \tag{65}$$

Under (59),

$$\delta M_A = -\frac{M}{2}\frac{D_A^{-1} - D_B^{-1}}{a^{-1} - L^{-1}}, \qquad \delta M_B = -\delta M_A. \tag{66}$$

For $D_A = D$, $D_B \gg D$, and $L \gg a$,

$$\delta M_A \simeq -\frac{Ma}{2D}, \qquad \delta M_B \simeq +\frac{Ma}{2D}. \tag{67}$$

Under the fixed-total convention, this is best described as a redistribution of exterior $1/r$ coefficients. In a relativistic support model, changes in shell energy, binding energy, or radiation can alter the total asymptotic mass budget.

F. Junction conditions and stability

For a cut-and-paste throat, the induced metric q_{ab} is continuous while a jump in extrinsic curvature is supported by a surface stress tensor:

$$\boxed{S_{ab} = -\frac{c^4}{8\pi G}\left([K_{ab}] - q_{ab}[K]\right).} \tag{68}$$

The junction conditions relate normal derivatives of the metric to the surface energy density and stresses [4], [5]. They do not supply an equation of state or guarantee that a quasistatic branch satisfying (60) is stable.

A passive gravitational conductor is particularly problematic. For a schematic split

$$M_A = \frac{\mathcal{M}}{2} + \delta, \qquad M_B = \frac{\mathcal{M}}{2} - \delta, \tag{69}$$

the δ-dependent gravitational self-energy has the generic form

$$E(\delta) = E(0) - C_{\mathrm{self}}\frac{G\delta^2}{a} + C_{\mathrm{int}}\frac{G\delta^2}{L} + \cdots, \tag{70}$$

with positive geometry-dependent coefficients. For $L \gg a$, the negative self-energy term dominates, so equal division is not a passive minimum. Any stable equalizing response requires exotic stresses, active control, or another constitutive mechanism.

VII. Time-Dependent Identification and Temporal Holonomy

The preceding sections assumed synchronized mouths. We now retain the spatial mouth map while allowing the two mouth time coordinates to acquire a relative offset.

A. Accumulated clock difference

Let the two mouth worldtubes be parametrized by a common internal label τ:

$$t_A = T_A(\tau), \qquad t_B = T_B(\tau). \tag{71}$$

The external time difference is

$$\Delta(\tau) = T_B(\tau) - T_A(\tau). \tag{72}$$

For a special-relativistic trip in which A remains inertial and τ is the proper time of B,

$$\Delta = \int (\gamma_B - 1)\,\mathrm{d}\tau = \int \left[1 - \sqrt{1 - \frac{v_B^2}{c^2}}\right] \mathrm{d}t. \tag{73}$$

At low speed,

$$\Delta \simeq \frac{1}{2c^2} \int v_B^2\,\mathrm{d}t. \tag{74}$$

In a weak gravitational field, the differential proper-time accumulation is

$$\tau_A - \tau_B \simeq \frac{1}{c^2} \int \left[\Phi_A - \Phi_B - \frac{v_A^2 - v_B^2}{2}\right] \mathrm{d}t. \tag{75}$$

The moving-mouth construction of a wormhole time machine is discussed by Morris, Thorne, and Yurtsever [2].

B. Final constant translation

After the acceleration ends, suppose both mouths are again static and their clock rates agree. The remaining identification can be

$$\boxed{(t, \mathbf{n})_A \sim (t + \Delta, P\mathbf{n})_B,} \tag{76}$$

with constant Δ. Because

$$\mathrm{d}(t + \Delta) = \mathrm{d}t, \tag{77}$$

a constant translation has the identity differential. It does not, by itself, alter the local induced metric, the local junction equations, or the tangent-space energy map. Frolov, Krtouš, and Zelnikov discuss constant time-gap identifications and distinguish them from dynamical situations in which a locally static but globally non-potential gravitational field makes the gap grow [11].

C. Holonomy in a one-exterior handle

The constant offset becomes invariant only when a path can traverse the throat and return through the same exterior, so that the comparison forms a closed loop. On a spatial manifold with a handle, choose a smooth closed one-form η satisfying

$$\mathrm{d}\eta = 0, \qquad \oint_\gamma \eta = 1, \tag{78}$$

where γ passes through the throat and returns externally. A stationary metric encoding the offset is

$$\boxed{\mathrm{d}s^2 = -N^2 c^2 (\mathrm{d}T - \Delta\eta)^2 + h_{ij}\mathrm{d}x^i \mathrm{d}x^j.} \tag{79}$$

Locally, $\eta = \mathrm{d}\chi$, so $\tau = T - \Delta\chi$ removes the cross term. Globally, no single-valued χ exists when η has nonzero period. The invariant statement is

$$\oint_\gamma \Delta\eta = \Delta. \tag{80}$$

Thus the time slip is locally removable but globally measurable by comparing the throat path with an exterior path.

VIII. Smooth Local Realization of a Constant Offset

The following Ellis-throat calculation does not, by itself, create nontrivial temporal holonomy. On a two-ended spacetime with disconnected asymptotic regions, it is a global change of clock convention. Its purpose is narrower: it shows that the local transition required by a prescribed asymptotic offset can be distributed smoothly through a finite region without adding curvature or a force on structureless test particles. The offset becomes holonomy only after the two ends are embedded or identified within a common exterior so that the relevant loop closes.

A. Ellis geometry and smooth clock interpolation

Consider the ultrastatic Ellis geometry [12], [13]:

$$\boxed{\mathrm{d}s_0^2 = -c^2 \mathrm{d}\tau^2 + \mathrm{d}l^2 + (l^2 + a^2)\mathrm{d}\Omega^2.} \tag{81}$$

The throat at $l = 0$ is smooth, and

$$R = -\frac{2a^2}{(l^2 + a^2)^2} \tag{82}$$

is finite everywhere.

Choose any smooth step function $F(l)$ such that, for some finite $L_F > 0$,

$$F(l) = 0 \quad (l \le -L_F), \qquad F(l) = 1 \quad (l \ge L_F). \tag{83}$$

Define

$$t = \tau + \Delta F(l). \tag{84}$$

Then

$$\boxed{\mathrm{d}s_\Delta^2 = -c^2[\mathrm{d}t - \Delta F'(l)\mathrm{d}l]^2 + \mathrm{d}l^2 + (l^2 + a^2)\mathrm{d}\Omega^2.} \tag{85}$$

The coordinate transition is confined to $|l| < L_F$.

B. No added local curvature

Equation (85) is obtained from the smooth single-valued transformation

$$\tau = t - \Delta F(l). \tag{86}$$

Consequently, all local curvature tensors and scalar invariants are unchanged:

$$R[g_\Delta] = R[g_0], \qquad R_{\mu\nu}R^{\mu\nu}[g_\Delta] = R_{\mu\nu}R^{\mu\nu}[g_0], \tag{87}$$

and similarly for the Riemann and Einstein tensors. There is no new thin shell or delta-function stress where $F' \neq 0$. In the two-ended Ellis manifold, the construction merely chooses asymptotic clocks whose zero points differ by Δ. In a completed one-exterior quotient, the same local chart can realize the transition associated with a separately imposed global holonomy.

C. Geodesics

Restrict to the equatorial plane. The geodesic Lagrangian is

$$2\mathcal{L} = -c^2(\dot{t} - \Delta F'\dot{l})^2 + \dot{l}^2 + (l^2 + a^2)\dot{\varphi}^2. \tag{88}$$

The conserved quantities are

$$E = c^2(\dot{t} - \Delta F'\dot{l}), \qquad J = (l^2 + a^2)\dot{\varphi}. \tag{89}$$

For a timelike geodesic parametrized by proper time,

$$\boxed{\dot{l}^2 = \frac{E^2}{c^2} - c^2 - \frac{J^2}{l^2 + a^2},} \tag{90}$$

$$\boxed{\ddot{l} = \frac{J^2 l}{(l^2 + a^2)^2}.} \tag{91}$$

Neither equation contains Δ. A radial geodesic has $J = 0$ and $\ddot{l} = 0$. The coordinate relation

$$t(\lambda) = \tau(\lambda) + \Delta F[l(\lambda)] \tag{92}$$

adds the asymptotic coordinate offset without changing the local four-momentum measured by corresponding static tetrads.

For the general static throat (6), the same transformation gives

$$\begin{aligned} \mathrm{d}s^2_\Delta = & -N^2(l)c^2[\mathrm{d}t - \Delta F'(l)\mathrm{d}l]^2 \\ & + \mathrm{d}l^2 + r^2(l)\mathrm{d}\Omega^2. \end{aligned} \tag{93}$$

With $\kappa = 1$ for timelike and 0 for null geodesics,

$$\dot{l}^2 = \frac{E^2}{N^2c^2} - \kappa c^2 - \frac{J^2}{r^2}, \tag{94}$$

$$\ddot{l} = -\frac{E^2 N'}{N^3 c^2} + \frac{J^2 r'}{r^3}. \tag{95}$$

The ordinary lapse force and angular-momentum barrier remain, but they are identical for $\Delta = 0$ and for constant $\Delta \neq 0$.

IX. Physical Effects of a Time Offset

A. Changing offsets

During acceleration, the temporal map can be

$$t_B = f(t_A) \tag{96}$$

rather than a translation. Then

$$\mathrm{d}t_B = f'(t_A)\mathrm{d}t_A, \tag{97}$$

and induced-metric matching requires a corresponding lapse relation. A changing offset can produce redshift, energy exchange, stresses, and radiation. When the final map becomes $f(t) = t + \Delta$, these rate-dependent effects vanish, although the accumulated offset remains.

B. Time-dependent fields and interference

For a field $\Psi(t, \mathbf{x})$, the boundary condition becomes

$$\Psi_A(t, \mathbf{n}) = \Psi_B(t + \Delta, P\mathbf{n}). \tag{98}$$

A Fourier component acquires phase

$$\widetilde{\Psi}_B(\omega) = e^{i\omega\Delta}\widetilde{\Psi}_A(\omega). \tag{99}$$

Static fields have $\omega = 0$ and are insensitive to a constant translation. Waves, oscillating fields, and interference measurements need not be.

C. Self-interaction and residual excitations

A structureless test particle neglects its own field and receives no additional local force from the constant offset in Section VIII. A charged or gravitating body can nevertheless send a field through the throat and interact with a delayed or advanced copy of that field. Such a force is nonlocal and depends on the global retarded Green function.

The acceleration history can also leave gravitational radiation, oscillations of the supporting fields, changes in mouth mass or spin, deformation, or stress in the accelerating apparatus. These are genuine local final-state data and are not determined solely by the number Δ.

D. Causal criterion

Let $\Delta_{\mathrm{adv}} > 0$ denote the time advance supplied in a chosen traversal direction. Let $T^{\min}_{\mathrm{wh}}$ be the minimum causal transit time through the finite throat in that convention, and $T^{\min}_{\mathrm{ext}}$ the minimum causal time for the exterior return leg. A closed causal curve becomes possible when

$$\Delta_{\mathrm{adv}} > T^{\min}_{\mathrm{wh}} + T^{\min}_{\mathrm{ext}}. \tag{100}$$

For a zero-length throat, $T^{\min}_{\mathrm{wh}} = 0$. This is a global causal effect, not a local force. A spacetime may remain locally smooth and stationary while failing to admit a global time function [2], [11].

X. Conclusions

The zero-length mouth model separates the differential of the boundary identification from its transverse extension. Tangential motion is fixed by the mouth map, but a complete momentum map requires an additional throat prescription. In the ideal elastic model, the transverse component reverses and the full map is Lorentz, so the mass shell is preserved. Momentum exchange with the mouths can be estimated in a common asymptotic frame, but its division among the two mouths, radiation, supports, and stored throat stress is constitutive.

Static scalar fields obey their ordinary local equations away from the throat. A chosen global-potential sector imposes pointwise potential continuity and opposite radial derivatives at corresponding mouths. Source-free throat flux is independent data. For two spherical flat exteriors with equal vanishing asymptotic potentials, the charge-generated same-side field is represented by a uniform line image ending at the Kelvin point. The resulting self-force is attractive in that convention. This construction is distinct from the single Kelvin point image of a grounded conductor.

The Newtonian scalar model has a controlled interpretation in weak exterior regions of general relativity. Under synchronized lapse matching, a monopole truncation fixes the antisymmetric local $1/r$ coefficient. An equal-and-opposite pair follows only after a fixed-total-monopole convention is imposed. The individual local coefficients near two mouths in one exterior are not separately ADM masses, and the stability and stored energy of the response require a throat action or controller.

A final constant clock offset has identity differential and therefore does not add a local force or change the local junction equations by itself. A smooth Ellis-type coordinate interpolation confirms that the local transition can be realized without new curvature. On a two-ended manifold this is only a choice of clock origins. It becomes invariant temporal holonomy when the ends participate in a common multiply connected exterior and the comparison path closes. The holonomy can then affect time-dependent fields, self-interaction, and global causality even though it remains locally removable.

Appendix A
Polyhedral Mouths

A. Facewise traversal map

Let a cubical mouth in its body frame be

$$C = \{\mathbf{y} : |y_x|, |y_y|, |y_z| \le a\}. \tag{101}$$

A global face identification can be represented by a signed permutation matrix P. On a face with outward body-frame normal $\mathbf{n}_f$, the ideal spatial map is

$$J_f = P(I - 2\mathbf{n}_f\mathbf{n}_f^T). \tag{102}$$

It maps tangential directions by P and reverses the face-normal component. In the external frame,

$$\mathbf{p}_{\rm out} = R_B J_f R_A^T \mathbf{p}_{\rm in}. \tag{103}$$

The map is constant across each open face. At an ideal edge or vertex more than one face normal is available, so the rule is ambiguous; a physical construction should round the edges or specify an edge-scattering law.

B. Curvature concentration

A transverse section through an edge of a cube exterior is a wedge of opening angle $3\pi/2$. Gluing two such exterior wedges produces a cone angle

$$\alpha_{\rm edge} = 3\pi, \tag{104}$$

with angular excess $-\pi$. The sharp three-dimensional quotient therefore has curvature concentrated along edges, with additional singular structure at vertices.

C. Distributional source mass

The angular excess at a sharp edge can be translated into an idealized line mass density. For a locally flat conical line source, let α be the total angle around the line and define the signed deficit

$$\delta = 2\pi - \alpha. \tag{105}$$

If λ denotes mass per unit proper length, the conical-defect relation is

$$\boxed{\delta = \frac{8\pi G}{c^2}\lambda, \qquad \lambda = \frac{c^2}{8\pi G}\delta.} \tag{106}$$

Thus an angular excess, $\delta < 0$, corresponds to a negative line mass density in this ideal stringlike description [3], [7]. For the cubical edge, $\alpha_{\rm edge} = 3\pi$, and hence

$$\delta_{\rm edge} = -\pi, \qquad \lambda_{\rm edge} = -\frac{c^2}{8G}. \tag{107}$$

The corresponding energy per unit length is $u_{\rm edge} = c^2\lambda_{\rm edge} = -c^4/(8G)$.

The cube in this appendix has side length $2a$, twelve edges, and total edge length $24a$. Neglecting any separately inserted point source at a vertex, the integrated edge mass is therefore

$$\boxed{M_{\rm cube}^{(\rm edge)} = 24a\lambda_{\rm edge} = -\frac{3ac^2}{G}.} \tag{108}$$

Numerically,

$$M_{\rm cube}^{(\rm edge)} \simeq -4.04 \times 10^{27} \left(\frac{a}{1\,\rm m}\right) \rm kg. \tag{109}$$

This value can also be obtained by rounding the polyhedron and then taking the sharp limit. For the symmetric gluing of two flat exteriors across a smooth static surface, the Israel conditions give the surface mass density

$$\Sigma_m = -\frac{c^2}{4\pi G}(\kappa_1 + \kappa_2), \tag{110}$$

where κ_1 and κ_2 are the principal curvatures with the convex-mouth sign convention [4]–[6]. Flat faces make no contribution. A rounded cube edge approaches a quarter-cylinder, whose integrated curvature reproduces Eq. (107); the integrated contribution of a rounded vertex vanishes with the rounding radius. A different vertex regularization may add a model-dependent point source and should then be included separately.

For comparison, gluing two flat spherical exteriors at radius a_s gives $\kappa_1 = \kappa_2 = 1/a_s$, so that

$$\Sigma_m^{(\rm sph)} = -\frac{c^2}{2\pi G a_s}, \qquad \boxed{M_{\rm sph} = 4\pi a_s^2 \Sigma_m^{(\rm sph)} = -\frac{2a_s c^2}{G}.} \tag{111}$$

Numerically,

$$M_{\rm sph} \simeq -2.69 \times 10^{27} \left(\frac{a_s}{1\,\rm m}\right) \rm kg. \tag{112}$$

Writing a_c for the cube half-side, the ratio of the magnitudes is

$$\boxed{\frac{|M_{\rm cube}^{(\rm edge)}|}{|M_{\rm sph}|} = \frac{3}{2}\frac{a_c}{a_s}.} \tag{113}$$

For equal parameters, $a_c = a_s$, the cubical construction carries $3/2$ times the integrated negative source mass of the spherical shell. Both estimates scale linearly with mouth size. They are proper source-mass integrals over the singular support, not ADM masses: because the two idealized exterior metrics are exactly flat near their asymptotic ends, the ADM mass at each end is zero.

D. Dirichlet–Neumann decomposition

Let G_D and G_N be the exterior-cube Dirichlet and Neumann Green functions, using the same radial normal convention on the cube boundary and decay at infinity. For a source on side A, define

$$\boxed{G_{AA} = \frac{1}{2}(G_N + G_D), \qquad G_{BA} = \frac{1}{2}(G_N - G_D).} \tag{114}$$

At the boundary, $G_D = 0$ and $\partial_\nu G_N = 0$, so the two functions have equal boundary values and opposite radial derivatives. This is the even/odd decomposition of the doubled exterior problem.

For a local infinite-plane approximation,

$$G_D = G_0 - G_{\rm im}, \qquad G_N = G_0 + G_{\rm im}, \tag{115}$$

so

$$G_{AA} = G_0, \qquad G_{BA} = G_{\rm im}. \tag{116}$$

The leading same-side planar image cancels. A finite cubical mouth responds through its finite extent, edges, corners, and global topology. No Kelvin inversion preserves a cube; the exact equivalent source is a continuous boundary density.

Near a conducting edge, the local transverse exterior wedge has opening angle $3\pi/2$. The leading Dirichlet harmonic behaves as $\rho^{\pi/(3\pi/2)} = \rho^{2/3}$, so its normal derivative and surface-charge density behave as

$$\sigma \sim \rho^{-1/3}. \tag{117}$$

The singularity is integrable. Standard wedge and conductor analyses are reviewed in [14], [15].

Appendix B
Illustrative Controlled Orbital Model

This appendix records a conservative toy controller. It is not derived from the preceding boundary conditions and is not claimed to be a passive wormhole response.

Let two remote fixed planets have equal mass M, and let a mouth of inertial mass m orbit each planet at radii r_1, r_2. For $L \gg a$ and negligible cross-potentials, the ideal zero-sum monopole formula gives

$$\mu = -\frac{Ma}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right). \tag{118}$$

Choose, as an explicit constitutive assumption, the positive stored energy

$$V_{\rm ctrl} = \frac{GM^2 a}{4}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)^2. \tag{119}$$

The effective Lagrangian is

$$L = \sum_{i=1}^{2}\left[\frac{m}{2}|\dot{\mathbf{r}}_i|^2 + \frac{GMm}{r_i}\right] - V_{\rm ctrl}. \tag{120}$$

The symmetric family $r_1(t) = r_2(t)$ has $\mu = 0$ and reduces to two ordinary Kepler problems.

For equal circular radii r_0, let $r_i = r_0 + x_i$, and define $x_- = x_1 - x_2$. Holding the unperturbed angular momenta fixed, the Kepler radial restoring frequency is GM/r_0^3, while

$$V_{\rm ctrl} = \frac{GM^2 a}{4r_0^4}x_-^2 + O(x^3). \tag{121}$$

The differential mode obeys

$$\ddot{x}_- + \omega_-^2 x_- = 0, \tag{122}$$

with

$$\boxed{\omega_-^2 = \frac{GM}{r_0^3}\left(1 + \frac{M}{m}\frac{a}{r_0}\right).} \tag{123}$$

This frequency is a consequence of the selected controller energy (119), not of wormhole topology. Varying a gravitational response coefficient while keeping the mouth's inertial mass fixed also raises an equivalence-principle consistency question that a complete relativistic support model would have to resolve.

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